

D825

Emerging Trends and Innovation in IT

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A research lab with a new cryogenic cooling system runs a 10-qubit processor. A reporter asks how many basis states the qubits can represent in superposition at once. The lab's grant is \$4 million. What is the answer?

- A. 20
- B. 1,024
- C. 100
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

How many qubits are needed to span 256 basis states?

- A. 16
- B. 128
- C. 8
- D. 256

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A hospital network with a new patient portal stores medical records that must stay confidential for 30 years, encrypted with RSA. A board member, a retired engineer, says quantum risk can wait until large quantum computers exist. What is the best response?

- A. Begin a cryptography inventory and plan migration to post-quantum algorithms now, since stolen data could be decrypted later.
- B. Double the RSA key length and consider it solved.
- C. Wait, because data is safe until quantum computers exist.
- D. Defer to the board member's engineering background.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which data is least urgent to protect with post-quantum cryptography?

- A.** Lifetime medical histories.
- B.** All data is equally urgent.
- C.** Government records classified for 50 years.
- D.** A one-time login code that expires in 60 seconds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A CIO at a regional grocery chain, who recently finished an executive program, proposes moving the email system and point-of-sale apps to a quantum computer to make them faster. What is the best assessment?

- A.** Poor fit because quantum is only useful for games.
- B.** Poor fit; quantum is expected to help with specific problems like molecular simulation or certain optimization, not everyday apps.
- C.** Good fit; quantum computers are faster at everything.
- D.** Good fit if the chain has at least 100 stores.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A technology is on magazine covers, vendors add it to every product, and few firms have it in production. An analyst firm with a famous logo calls it “transformational.” Where on the hype cycle is it most likely?

- A.** Plateau of productivity
- B.** Trough of disillusionment
- C.** Peak of inflated expectations
- D.** It is mature because an analyst praised it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

After a wave of failed pilots and negative headlines, a technology is being quietly used for a few practical, well-defined tasks. Where is it most likely?

- A.** Slope of enlightenment
- B.** It has failed and will disappear.
- C.** Peak of inflated expectations
- D.** Innovation trigger

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An oil company with a well-known safety record runs sensors on offshore rigs that must shut valves within 10 milliseconds of a pressure spike. The satellite link to the cloud adds about 600 milliseconds. Where should the shutdown logic run?

- A.** On edge devices at the rig.
- B.** In the cloud, for more computing power.
- C.** In the cloud with a faster server.
- D.** It does not matter where logic runs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A fintech startup with a bright orange brand finds security flaws only in a two-week review before each release, delaying launches. It deploys code 20 times a month. What change fits DevSecOps?

- A.** Skip security to ship faster.
- B.** Hire more testers for the final review.
- C.** Extend the final review to four weeks.
- D.** Build automated security scans into the pipeline so flaws are caught at each commit.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A smart-glasses startup sold out to tech enthusiasts at launch, helped by a viral video. Sales to mid-size businesses have stalled because buyers want proven references and integration support. What does this show?

- A. Another viral video will win the mainstream.
- B. Success with innovators guarantees mainstream success.
- C. The product has reached the late majority.
- D. The product is stuck at the chasm; the early majority wants proven, practical solutions.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which buyer group values novelty most and tolerates bugs?

- A. Laggards
- B. Early majority
- C. Innovators
- D. Late majority

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A logistics firm's CIO, who keeps a popular tech blog, sees growing talk of autonomous delivery robots but high uncertainty about regulation and cost. What approach best fits visionary IT leadership?

- A. Ignore the trend until competitors adopt it.
- B. Build scenarios and fund a small pilot with clear learning goals before deciding to scale.
- C. Write more blog posts about it.
- D. Replace all drivers with robots next year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A research lab ~~with a new cryogenic cooling system~~ runs a 10-qubit processor. A reporter asks how many basis states the qubits can represent in superposition at once. ~~The lab's grant is \$4 million.~~ What is the answer?

Target word: number of states **Decides:** 2^{10} .

A	Inverted operation	2×10 multiplies instead of raising to a power.
B	Correct	$2^{10} = 1,024$.
C	Confusion set	10^2 swaps base and exponent.
D	Wrong stem number	The qubit count.

2. Answer C Concept 1 • transfer

How many qubits are needed to span 256 basis states?

Target word: solve for qubits **Decides:** $2^n = 256$.

A	Confusion set	$16^2 = 256$ swaps base and exponent.
B	Inverted operation	$256 / 2$ uses the $2n$ model.
C	Correct	$2^8 = 256$.
D	Wrong stem number	Repeats the state count.

3. Answer A Concept 2

A hospital network ~~with a new patient portal~~ stores medical records that must stay confidential for 30 years, encrypted with RSA. A board member, ~~a retired engineer~~, says quantum risk can wait until large quantum computers exist. What is the best response?

Target word: timing of quantum risk **Decides:** Long-lived data + harvest now, decrypt later.

A	Correct	Long confidentiality needs make the risk current.
B	Dropped element	Shor's algorithm undermines RSA at scale regardless.
C	Inverted operation	Ignores harvest-now, decrypt-later.
D	Camouflage	Background does not change the threat.

4. Answer D Concept 2 • transfer

Which data is least urgent to protect with post-quantum cryptography?

Target word: reverse: least urgent **Decides:** Short-lived data loses value before quantum risk arrives.

A	Confusion set	Long-lived and sensitive.
B	Absolute language	Urgency depends on how long data must stay secret.
C	Inverted operation	The most urgent case.
D	Correct	Worthless long before any quantum decryption.

5. Answer B Concept 3

A CIO at a regional grocery chain, ~~who recently finished an executive program~~, proposes moving the email system and point-of-sale apps to a quantum computer to make them faster. What is the best assessment?

Target word: fit for quantum **Decides:** Routine workloads are not quantum problems.

A	Confusion set	Misstates where quantum helps.
B	Correct	Quantum advantage is problem-specific.
C	Absolute language	Overgeneralizes quantum speedup.
D	Wrong stem number	Scale does not change the kind of problem.

6. Answer C Concept 4

A technology is on magazine covers, vendors add it to every product, and few firms have it in production. An analyst firm ~~with a famous logo~~ calls it “transformational.” Where on the hype cycle is it most likely?

Target word: hype cycle position **Decides:** Buzz with little production use.

A	Inverted operation	Plateau means broad, proven use.
B	Confusion set	The trough follows failed pilots and fading buzz.
C	Correct	High hype, low proven use.
D	Camouflage	Praise does not show production maturity.

7. Answer A Concept 4 · transfer

After a wave of failed pilots and negative headlines, a technology is being quietly used for a few practical, well-defined tasks. Where is it most likely?

Target word: later position **Decides:** Past the trough, practical uses emerging.

A	Correct	Practical uses emerge after the trough.
B	Absolute language	Negative headlines often precede real value.
C	Inverted operation	Buzz has already faded.
D	Confusion set	The very first stage.

8. Answer A Concept 5

An oil company ~~with a well-known safety record~~ runs sensors on offshore rigs that must shut valves within 10 milliseconds of a pressure spike. The satellite link to the cloud adds about 600 milliseconds. Where should the shutdown logic run?

Target word: edge vs. cloud **Decides:** Latency far exceeds the response limit.

A	Correct	Only local processing meets a 10 ms limit.
B	Inverted operation	600 ms round trip misses the limit.
C	Dropped element	Server speed does not remove link latency.
D	Absolute language	Latency matters in real-time control.

9. Answer D Concept 6

A fintech startup ~~with a bright orange brand~~ finds security flaws only in a two-week review before each release, delaying launches. It deploys code 20 times a month. What change fits DevSecOps?

Target word: DevSecOps practice **Decides:** Security moved into the pipeline early.

A	Inverted operation	Trades away security.
B	Adjacent category	Still security at the end.
C	Wrong stem number	A longer late gate slows delivery further.
D	Correct	Shifts security left and removes the late gate.

10. Answer D Concept 7

A smart-glasses startup sold out to tech enthusiasts at launch, ~~helped by a viral video~~. Sales to mid-size businesses have stalled because buyers want proven references and integration support. What does this show?

Target word: adoption lifecycle **Decides:** Early adopters vs. early majority needs.

A	Camouflage	Buzz does not meet pragmatic needs.
B	Absolute language	The chasm exists because it does not.
C	Confusion set	It has not yet reached the early majority.
D	Correct	Mainstream buyers need references and support.

11. Answer C Concept 7 · transfer

Which buyer group values novelty most and tolerates bugs?

Target word: reverse identification **Decides:** Innovators.

A	Confusion set	Adopt last and resist change.
B	Inverted operation	Want proven, reliable solutions.
C	Correct	Seek novelty and accept rough edges.
D	Adjacent category	Skeptical and cost-conscious.

12. Answer B Concept 8

A logistics firm's CIO, ~~who keeps a popular tech blog~~, sees growing talk of autonomous delivery robots but high uncertainty about regulation and cost. What approach best fits visionary IT leadership?

Target word: leading under uncertainty **Decides:** Scan, scenario-plan, pilot small.

A	Inverted operation	Risks falling behind.
B	Correct	Small bets limit risk while building knowledge.
C	Camouflage	Commentary is not strategy.
D	Absolute language	An all-in bet under high uncertainty.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Qubits and superposition	A qubit can hold a superposition of 0 and 1; n qubits together span 2^n basis states.	Thinking n qubits hold $2n$ states, or that a qubit is just a faster bit.	A qubit count and a question about how many states.
2	Quantum risk to public-key cryptography	A large fault-tolerant quantum computer running Shor's algorithm could break RSA and elliptic-curve encryption, so organizations should inventory crypto and plan migration to post-quantum algorithms now, because data stolen today can be decrypted later.	Assuming there is nothing to do until quantum computers exist.	Long-lived sensitive data protected by RSA or ECC.
3	Quantum advantage is problem-specific	Quantum computers are expected to excel at specific problems (simulation, certain optimization, factoring), not to replace classical computers for general tasks.	Expecting quantum to speed up email, web apps, or everyday databases.	A proposal to move routine workloads to quantum.
4	The hype cycle	New technologies often go from a trigger to a peak of inflated expectations, a trough of disillusionment, a slope of enlightenment, and a plateau of productivity.	Reading a peak of buzz as proof of maturity, or a trough as proof of failure.	Heavy media buzz with few production deployments, or a wave of failed pilots.
5	Edge computing reduces latency	Edge computing processes data near where it is created, cutting latency and bandwidth when real-time response matters.	Sending all data to a distant cloud even when milliseconds matter.	Real-time decisions at remote or mobile sites.
6	DevSecOps shifts security left	DevSecOps builds automated security testing into the delivery pipeline from the start instead of a final gate.	Adding a security review only before release.	Late-found vulnerabilities delaying releases.
7	Technology adoption lifecycle	Adopters move from innovators to early adopters, early majority, late majority, and laggards; the early majority wants proven, practical solutions (the chasm).	Assuming what wins early adopters will win the mainstream.	Strong early interest that stalls in mainstream sales.
8	Visionary leaders place small bets and scan the horizon	Visionary IT leaders scan emerging trends, use scenario planning, and fund small pilots with clear learning goals before scaling.	Betting everything on one trend, or ignoring trends until competitors act.	An uncertain technology with potential strategic impact.

Course scope (public description): Emerging Trends and Innovations in IT offers a forward-looking perspective introducing learners to upcoming technological innovations from quantum mechanics' role in computing to the potential of advanced AI. Emphasis is placed on the evolving methodologies in information technology (IT) management and the attributes required for visionary IT leadership in anticipating future tech landscapes.